

Jackson Ferguson

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Education

University of Victoria

Victoria, BC

B.Sc. Major Combined Physics and Astronomy

Graduated Spring 2026

High Performance Coursework: Quantum Computing (100%), Stellar Astrophysics (99%), Probability & Statistics (97%), Computer Assisted Physics (91%), Experimental Physics (88%), Thermodynamics (86%).

Technical Skills

Languages: Python 3.10-3.14, Bash, TOML, YAML, $\LaTeX$
Software Architecture: CLI Design, Transactional Filesystems, AST Parsing (tomlkit), Git Internals
Data & Parsing: Pandas, NumPy, SciPy, pyparsing, pdfplumber, Jupyter
DevOps & Distribution: Git/GitHub Actions, Docker/GHCR, uv, PyPI, Homebrew
Testing & QA: Pytest, Hypothesis, Snapshot Testing, Mypy (Strict), Ruff, Bats/ShellCheck

Development & Research Projects

Protostar: Managed Python Scaffolding & Reconciliation Engine *Python, State Reconciliation, CLI Architecture*

- **Managed Project Lifecycle:** Built a recipe- and ownership-driven lifecycle (status, diff, sync) that reviews tooling and template changes against live repositories, applies safe updates, preserves local edits, and advances baselines deterministically.
- **Semantic Reconciliation:** Implemented three-way and semantic reconciliation for TOML, YAML, GitHub Actions, dependencies, and generated text, preserving foreign content while surfacing key- or line-level conflicts.
- **Transactional Execution:** Separated read-only planning from mutation and added stale-input revalidation, mutation journaling, subprocess teardown, and byte/permission rollback on failures or SIGINT.
- **Multi-Interface Architecture:** Designed one orchestration core behind Textual review flows, headless CLI, dry-run, and structured JSON, including versioned repository-template upgrades and migrations.

CI/CD & Release Infrastructure

GitHub Actions, Python, Release Automation

- **Two-Phase Release Orchestration:** Implemented read-only preflight checks for branch, cleanliness, remote synchronization, version metadata, and tag collisions before any release state is mutated.
- **Transactional Publication:** Automated version/lock updates, release commits, annotated tags, and git push --atomic, with local rollback when execution fails or is interrupted.
- **Reusable Distribution Pipelines:** Built centralized workflows that wait for PyPI artifacts, resolve dependency trees with uv, generate Homebrew resources, and audit updated formulae for caller repositories.

Star Ground: Deterministic Hardware Data Pipeline

Python, pyparsing, Streamlit, Docker

- **Structured Ingestion:** Unified PDF, CSV, pasted text, URLs, and presets behind parser strategies; implemented a grammar that converts engineering notation such as 4k7 into exact Decimal values.
- **Reliability Testing:** Combined golden-master regression snapshots, Hypothesis fuzzing of malformed inputs and mathematical invariants, and headless Streamlit tests for end-to-end verification.

Dark Matter: Dependency-Graph Storage Profiler

Python, Graph Traversal, Homebrew

- **Fractional Dependency Attribution:** Reconstructed Homebrew's dependency DAG and computed weighted recursive storage costs that divide shared dependencies across all parents instead of double-counting them.
- **Measurement-Aware Analysis:** Separated exact installed-size scans from theoretical catalog analysis, exposing the approximation introduced by compressed bottle sizes while preserving useful relative comparisons.

Galaxy Cluster Dark Matter Analysis (ACO 2670)

Astrophysics Research Project

- **Pipeline Development:** Constructed a data pipeline to clean spectroscopic data for 98+ galaxies, utilizing redshift-space distortion analysis to isolate cluster members from the background field.
- **Statistical Modeling:** Applied the Virial Theorem and Gaussian Mixture Models (GMM) to calculate a mass-to-light ratio of $(291 \pm 60) M_{\odot}/L_{\odot}$, providing kinematic evidence of Dark Matter dominance.

Work & Leadership Experience

Co-Lead Event Organizer, Unplugged Events, Victoria, BC

Jan 2025 – Jan 2026

- Managed technical production, signal flow, and venue logistics for live events (100+ attendees).

Server, Stanley Park Brewing, Vancouver, BC

May 2025 – August 2025

Alpine Ski Instructor, Whistler Blackcomb, Whistler, BC

Nov 2018 – Feb 2020

Freelance 3D Modeler, Twisted Time Studios, Remote

May 2015 – Aug 2015

- Designed and created main character models and weapon systems for Extinction: Archaic Genesis.